



Multi-Decadal Land-Cover Mapping of the Chittagong Hill Tracts Using Machine Learning and Spatial Validation

A thesis submitted in partial
fulfillment of the requirements for the
degree of

Bachelor of computer science and Engineering

By

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Session: 2017 – 2018

Reg: 2017 – 14 – 55

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I, **Shohinur Pervez Shohan**, do hereby declare that this report has been done by me under the supervision of **Dhiman Sarma**, Associate Professor, Department of Computer Science and Engineering, Rangamati Science and Technology University. I also declare that this report, nor any part of this report, has been submitted elsewhere for the award of any degree.

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ACKNOWLEDGEMENT

Completing a thesis requires support from many individuals. I am deeply thankful to all those who contributed to this endeavor. I convey my sincere gratitude to my honorable Supervisor, **Dhiman Sarma**, Associate Professor of the Department of Computer Science and Engineering, Rangamati Science and Technology University. Without his guidance and support, this study would not have reached its successful completion. His invaluable supervision and direction have greatly shaped the outcome of this report.

I am also grateful to all my teachers in the Department of Computer Science and Engineering, Rangamati Science and Technology University. I extend my appreciation to my batchmates **Mohammed Mujibur Rahman Shourav** (Registration: 2017-14-17) and **Najmul Hasan Shanto** (Registration: 2017-14-20) for their support and encouragement throughout this journey.

My heartfelt gratitude goes to my family for their unwavering support and encouragement. This thesis is dedicated to my father **Md. Fazlul Karim**, my mother **Miss Rozi Akter**, my wife **Fatiha-Tun-Noor Binte Saif** and my family who have always been there for me.

Finally, I would like to thank everyone who played a part, however big or small, in making this thesis a reality. I am forever indebted to their support and am honored to have had them by my side throughout this journey.

Shohinur Pervez Shohan

Abstract

Monitoring land-cover change in tropical hill landscapes requires distinguishing net forest extent from shifts among operational forest classes. This study evaluated land-cover and forest-composition change in Rangamati District, Bangladesh, using six Landsat epochs: 1993, 1998, 2003, 2008, 2018, and 2023. Google Earth Engine was used to prepare dry-season Landsat Collection 2 surface-reflectance composites, harmonize Landsat 8 data to the Landsat 5/7 spectral domain, and derive spectral and terrain predictors. Five operational classes were mapped: Dense Forest, Degraded Forest/Jhum, Water, Agriculture/Settlement, and Bare Land. Random Forest, support vector machine, and classification and regression tree classifiers were compared with a spatial-block holdout; Random Forest was then used for production mapping. The 2023 map was evaluated with 200 manually interpreted reference points and area-adjusted estimation, yielding an overall accuracy of 73.17% (95% confidence interval: 64.34–82.00%). Mapped combined forest area increased by 20,693 ha, or 4.4% relative to the 1993 baseline, whereas Dense Forest decreased by 8,235 ha. Sensitivity analysis showed that changing the normalized difference vegetation index threshold altered Dense Forest area by 27% but combined forest area by only 0.7%. The marked redistribution between Dense Forest and Degraded Forest/Jhum during 2003–2008–2018 remained unresolved and was not interpreted as confirmed forest loss or recovery. Area-adjusted validation was limited to the 2023 map; historical classifications and transition estimates were not subjected to equivalent reference-point validation. The results provide a documented district-scale basis for monitoring land-cover change while showing that net forest area alone does not capture the uncertainty and redistribution associated with operational forest classes.

1. Introduction

Forest-change monitoring in tropical mountain regions must distinguish changes in forest extent from changes in forest condition. Forest conversion denotes a transition from forest to another land-cover type, whereas degradation can involve reduced canopy cover, altered structure, fragmentation, or declining biomass without complete removal of tree cover. Regeneration adds a further complication because recovering vegetation may share spectral characteristics with other covers at particular stages of succession. These distinctions matter for landscape planning and for interpreting satellite-derived maps: a stable aggregate forest area can coexist with substantial redistribution among closed forest, secondary vegetation, cultivation fallows, and non-forest land. Yet degradation is more difficult to define and observe consistently than deforestation, particularly with medium-resolution optical imagery in structurally complex tropical landscapes (Dupuis et al., 2020).

The Chittagong Hill Tracts (CHT) of southeastern Bangladesh provide a demanding setting in which to examine these issues. The region comprises the hill districts of Rangamati, Khagrachhari, and Bandarban, where steep terrain and monsoonal seasonality coincide with mosaics of natural and planted forest, shifting cultivation (jhum), agricultural land, settlements, bare or sparsely vegetated surfaces, and water bodies. The Hill Tracts hold roughly 40% of Bangladesh's total forest land area while occupying only about 8% of its territory, so a district-scale record here carries

national relevance for forest monitoring. Rangamati, the largest of the three hill districts, includes the Kaptai Reservoir, for which changing water levels can further complicate interpretation of mapped land-cover transitions. Jhum cultivation and secondary vegetation are especially relevant because clearing, cultivation, fallow, and regrowth can produce short-lived and spectrally overlapping land-cover states. Accordingly, a change in a broad “forest” class cannot by itself establish either forest loss or ecological recovery in this landscape.

Landsat provides an unusually valuable record for investigating such multi-decadal dynamics at district scale. Its long archive enables comparison across decades at a spatial resolution compatible with regional land-cover mapping, while cloud-based processing platforms reduce the computational barrier to assembling and analysing multi-sensor image collections. However, a long satellite record does not automatically yield directly comparable maps. Sensor transitions, atmospheric conditions, residual clouds, seasonal phenology, terrain effects, compositing choices, and mixed pixels can alter spectral responses between observation dates. The distinction between dense forest and degraded, secondary, or shifting-cultivation vegetation is therefore particularly sensitive to class definitions and reference-label quality. Inter-sensor harmonization is an important safeguard for analyses spanning Landsat 5, Landsat 7, and Landsat 8, but it cannot remove all sources of temporal classification uncertainty.

Recent work demonstrates both the value and the limits of land-cover mapping in Bangladesh and comparable tropical landscapes. At the CHT scale, Chakma et al. (2023) mapped 1998, 2008, and 2018 Landsat imagery using object-based classification and a four-class legend comprising forest, grassland, water bodies, and bare land. This provides an important regional multi-temporal reference, but its aggregated forest class does not resolve differences between denser woody cover and degraded or transitional vegetation. At a more localized Bangladesh site, Hossain et al. (2023) applied supervised maximum-likelihood classification to Landsat imagery to map land-cover change around Fashiakhali Wildlife Sanctuary from 1994 to 2021. Together, these studies show that mapped trajectories vary by spatial scale, class scheme, time period, and analytical design. They also demonstrate why a district-wide result should not be treated as a substitute for evidence from protected areas or other individual landscapes.

Methodological choices further affect how land-cover change can be interpreted. Comparisons of classifiers are common, but random train-test partitions can give overly optimistic performance estimates when neighbouring observations are spectrally and spatially dependent. In addition, conventional confusion-matrix summaries do not by themselves convert a stratified reference sample into unbiased area estimates. Design-based area adjustment makes the mapped-area weights explicit and supports uncertainty intervals for class areas and accuracy estimates. Reference-label quality remains a separate concern, especially where global products are used to construct training data. For example, ESA WorldCover supplies a valuable spatially consistent source of labels, but independent comparison has shown meaningful local disagreement between global and national land-cover products (Duarte et al., 2023). These considerations are central in data-scarce settings, where apparently precise change estimates can exceed the strength of their underlying reference information.

The resulting research gap is not an absence of land-cover research in the CHT, but a need for a transparent district-scale record that combines multi-decadal coverage with an explicit operational

distinction between Dense Forest and Degraded Forest/Jhum and proportionate treatment of uncertainty. Previous studies differ in temporal coverage, class aggregation, image-processing workflow, and validation approach, making direct comparison of reported forest trends difficult. In particular, broad forest categories may conceal exchange between denser forest and degraded or transitional cover, whereas validation based on spatially dependent samples or unadjusted mapped areas can overstate certainty. A study of Rangamati should therefore distinguish operational map classes from field-measured ecological condition, evaluate classification performance under spatial separation, and avoid treating historical transitions as independently validated facts where contemporaneous reference evidence is unavailable.

This study aims to reconstruct multi-temporal land-cover and forest-composition change across Rangamati District between 1993 and 2023 and to interpret it with appropriate treatment of uncertainty. Specifically, it: (i) maps five land-cover classes for six Landsat epochs, 1993, 1998, 2003, 2008, 2018, and 2023; (ii) compares Random Forest (RF), support vector machine (SVM), and classification and regression tree (CART) classifiers using a spatial-block holdout; (iii) assesses the 2023 RF map using manually interpreted reference observations and area-adjusted accuracy and area estimation; (iv) examines the sensitivity of estimated forest extent to the operational normalized difference vegetation index (NDVI) threshold used to distinguish Dense Forest from Degraded Forest/Jhum; and (v) quantifies mapped transitions, separating forest–non-forest conversion from exchange between the two forest-related classes.

The study addresses five related questions: How did mapped Dense Forest, Degraded Forest/Jhum, and non-forest classes vary across the six epochs? How did RF, SVM, and CART perform under a spatial-block holdout? How do the manually interpreted reference assessment and area adjustment qualify interpretation of the 2023 map? How sensitive is the mapped forest extent to the Dense Forest definition? Finally, do mapped transitions primarily indicate forest–non-forest conversion or redistribution within the forest-related classes? The contribution is an integrated and documented application using openly available processing tools, combining Landsat surface-reflectance processing, sensor harmonization, classifier comparison, spatially separated evaluation, contemporary area-adjusted assessment, sensitivity analysis, and transition mapping for a data-scarce hill district. It does not claim independent validation of the historical maps or causal attribution of observed change. Rather, it provides an evidence-bounded basis for interpreting a multi-decadal land-cover record and for identifying where stronger historical reference data would be most valuable.

2. Materials and Methods

2.1 Study area

The investigation centred on Rangamati District, one of the three hill districts that make up the Chittagong Hill Tracts in southeastern Bangladesh, and encompassed the full administrative extent of the district, covering all ten upazilas. The Chittagong Hill Tracts experience rapid, spatially heterogeneous land-cover change driven by the expansion of permanent agriculture at the expense of shifting cultivation (jhum) and forest, a dynamic common to tropical hillslope systems worldwide. The administrative-boundary shapefile was obtained from DIVA-GIS (<https://www.diva-gis.org/>), which distributes country-level administrative boundaries at multiple

levels for free download. The Bangladesh Level-2 administrative boundary layer was downloaded and uploaded to Google Earth Engine as a private table asset (BGD_adm2). The study area's region of interest (ROI) was then derived by selecting the polygon whose NAME_2 attribute is recorded as "Parbattya Chattagram". That value is a naming artifact on the source shapefile rather than a scope change; it persists as a legacy descriptor, and the corresponding polygon was therefore taken to represent the Rangamati District boundary throughout the study. The same ROI governed every subsequent operation, from image compositing and training-sample extraction through classification, validation, area estimation and the multi-temporal transition analysis.

The study area is illustrated in Figure 1.

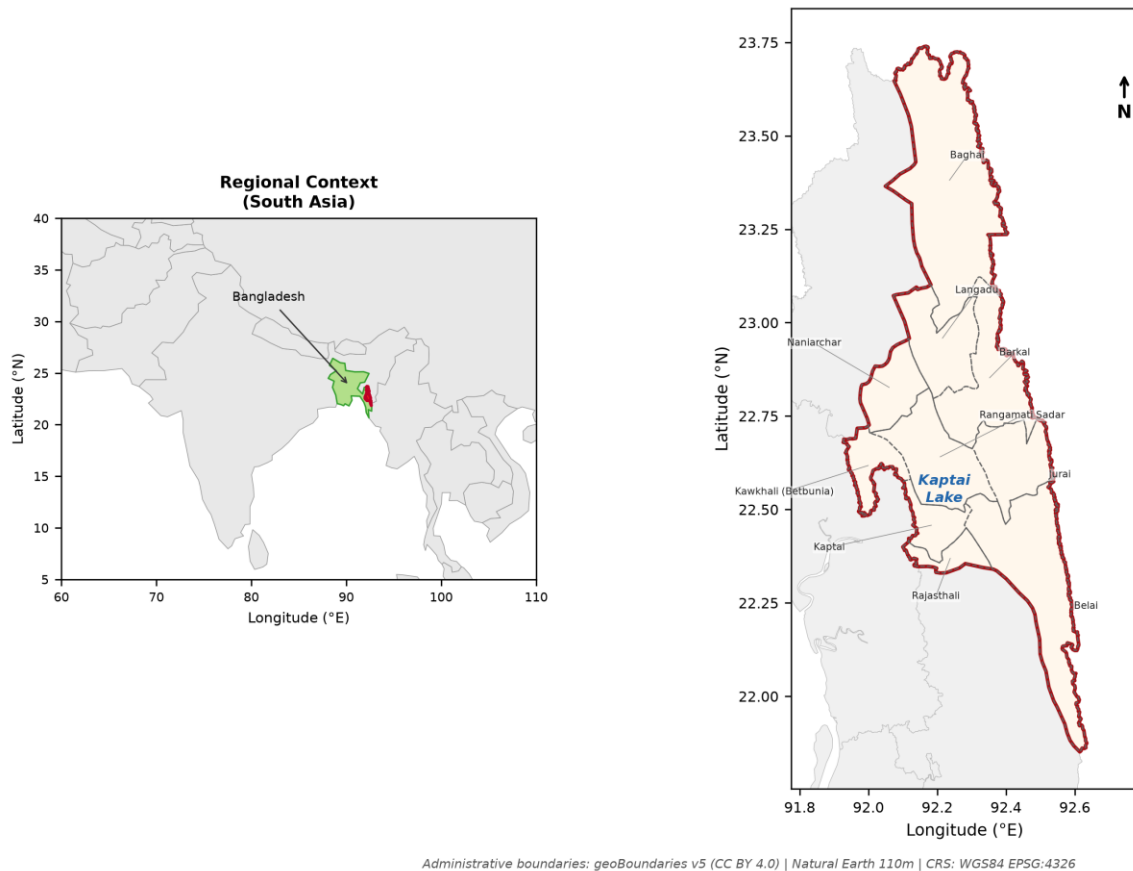


Figure 1. Study area. Left: regional context of Rangamati District within Bangladesh and South Asia. Right: the Rangamati District boundary with upazila divisions and Kaptai Lake. Base data from geoBoundaries and Natural Earth.

2.2 Satellite and ancillary data

Land-cover dynamics over the 1993–2023 period were reconstructed from a six-epoch Landsat time series designed to capture three decades of change. Dense, multi-sensor Landsat time series have become a standard basis for decadal forest and land-cover mapping. Landsat 5 Thematic Mapper (TM) data supplied the 1993, 1998 and 2008 epochs; Landsat 7 Enhanced Thematic

Mapper Plus (ETM+) provided the 2003 epoch; and Landsat 8 Operational Land Imager (OLI) imagery was used for 2018 and 2023. All scenes were drawn from the USGS Landsat Collection 2, Level 2 surface-reflectance archive (U.S. Geological Survey, 2022). To minimise phenological and moisture-driven variability associated with the monsoon, acquisitions were restricted to the 1 January–30 April window in each observation year. A 2013 Landsat 8 epoch was compiled during preliminary processing but excluded from the final classification and change analysis, as the available observations failed to yield a sufficiently cloud-free composite.

Training labels were derived from the ESA WorldCover version 200 (2021; Zanaga et al., 2022) reference map. WorldCover is a global 10 m product whose published overall accuracy is in the region of 75%, with notable local variability, so its labels should be treated as a practical training/reference source rather than as ground truth. Because WorldCover depicts approximately the 2021 land-cover condition, a training location whose class changed before or after that year carries a label that may not match its condition in a given historical epoch. The historical maps should therefore be understood as model-based reconstructions produced under a common reference-label framework, not as independently validated historical land-cover maps. This temporal mismatch is most consequential for the earliest epochs examined here. Elevation was sourced from the 30 m Shuttle Radar Topography Mission product (USGS/SRTMGL1_003; Farr et al., 2007), and slope was computed from the elevation layer using Google Earth Engine (Gorelick et al., 2017) terrain functions.

2.3 Image pre-processing and compositing

Each scene first underwent quality-based masking. Pixels flagged in the QA_PIXEL band as dilated cloud, cloud, cloud shadow or snow were discarded, and radiometrically saturated pixels were removed via QA_RADSAT. Surface reflectance was then recovered from the raw digital numbers using the Collection 2 scaling relation (Eq. 1).

$$\rho_{\lambda} = 0.0000275 \times DN_{\lambda} - 0.2 \quad (1)$$

Rather than applying a scene-level CLOUD_COVER threshold, the workflow controlled contamination at the pixel level through quality masking combined with a median composite computed over the dry-season window. To ensure spectral comparability across sensors, Landsat 5 and Landsat 7 imagery were first mapped to a common band set (Blue, Green, Red, NIR, SWIR1 and SWIR2); Landsat 8 reflectance was then transformed into the Landsat 5/7 spectral space using the OLI-to-ETM+ harmonization coefficients of Roy et al. (2016), following the inter-sensor harmonization practice advocated for multi-decadal Landsat forest mapping.

Four spectral indices were derived for each composite (Eqs. 2–5): the Normalized Difference Vegetation Index (NDVI), which tracks vegetation greenness; the Normalized Difference Water Index (NDWI), employed to support water delineation; the Enhanced Vegetation Index (EVI), selected for its reduced sensitivity to atmospheric effects and canopy background; and the Normalized Burn Ratio (NBR), which captures vegetation and moisture contrast (Kaufman & Remer, 1994).

$$NDVI = (NIR - Red) / (NIR + Red) \quad (2)$$

$$NDWI = (Green - NIR) / (Green + NIR) \quad (3)$$

$$EVI = 2.5 \times (NIR - Red) / (NIR + 6 \times Red - 7.5 \times Blue + 1) \quad (4)$$

$$NBR = (NIR - SWIR2) / (NIR + SWIR2) \quad (5)$$

This yielded a twelve-variable feature space per epoch, comprising the six surface-reflectance bands, the four indices, and the elevation and slope layers. Because terrain is effectively static over the study period, these two layers were treated as time-invariant and appended to the feature stack in every epoch. The constituent bands entered the analysis at their native resolutions: the Landsat-derived spectral indices at 30 m and the SRTM-derived elevation and slope at approximately 30.87 m. The layers were combined in Earth Engine through `addBands()` without an explicit reprojection step; the platform resampled all inputs to the common 30 m analysis grid at the point of sampling, using its default nearest-neighbour resampling via the `scale: 30` parameter passed to every stratified sampling and area-reduction call.

2.4 Land-cover classification scheme

Land cover was mapped into five classes derived from the ESA WorldCover legend, with the tree-cover category subdivided on the basis of NDVI. The class definitions were fixed prior to the multi-temporal analysis and are summarised in Table 1.

Table 1. *Operational definitions of the five mapped land-cover classes*

Class	Operational definition
1. Dense Forest	WorldCover tree cover (class 10) and $NDVI \geq 0.55$
2. Degraded Forest/Jhum	WorldCover tree cover and $0.25 \leq NDVI < 0.55$, or WorldCover shrubland (20) or grassland (30)
3. Water	WorldCover permanent water bodies (class 80)
4. Agriculture/Settlement	WorldCover cropland (40) or built-up (50)
5. Bare Land	WorldCover bare or sparse vegetation (class 60)

An NDVI threshold of 0.55 provided the primary operational boundary between Dense Forest and Degraded Forest/Jhum. The cutoff should be understood as a decision rule rather than a direct field measurement of canopy condition, and its influence on the estimated forest extent is examined separately in Section 2.10. Training labels were derived from a single fixed 2023 NDVI composite, applied uniformly as an NDVI threshold within the ESA WorldCover 2021 tree-cover pixels to distinguish Dense Forest from Degraded Forest/Jhum. This same 2023-NDVI-based labelling scheme was used consistently across training-sample generation, area calculation, and the transition analysis. Independent validation labels were instead assigned through visual interpretation of high-resolution imagery (Section 2.8), and the forest-definition sensitivity analysis (Section 2.10) deliberately varied the NDVI threshold rather than holding it fixed. The Degraded Forest/Jhum class is intended to capture the transitional cover typical of shifting-cultivation landscapes, which remain widespread across tropical hillslopes and are notoriously

difficult to map from satellite imagery because of their rapid spectral turnover (cf. Saleh et al., 2019).

2.5 Training-sample generation

Training data were obtained through stratified random sampling from the five-class reference map, with a nominal allocation of 150 samples per class yielding a target of 750 locations. The random seed was fixed at 42 to guarantee reproducibility, and feature vectors were extracted from the twelve-band composite at each sample location. For production mapping, a single Random Forest model was trained on the 2023 composite and applied unchanged to every epoch's composite, so that the historical maps are produced by temporal transfer of one fixed 2023-trained classifier rather than by epoch-specific retraining. This is the basis for treating the historical classifications as model-based reconstructions under a common reference-label framework (Section 2.2).

2.6 Spatial-block holdout design

To limit the confounding effect of spatial autocorrelation, training and holdout observations were partitioned geographically rather than at random. This design addresses a well-documented pitfall whereby spatially dependent training and validation samples inflate reported accuracy, and spatially blocked splits are a widely recommended remedy for data with spatial structure (Karasiak et al., 2021; Lyons et al., 2018; Ramezan et al., 2019). We acknowledge that spatially structured cross-validation is not universally favoured for map-accuracy inference, where design-based estimators are sometimes argued to be the more defensible target; here, however, the spatial-block holdout was chosen to provide a conservative, spatially independent assessment of classifier generalization in a region of strong local autocorrelation. Each sample was assigned to a $0.1^\circ \times 0.1^\circ$ longitude–latitude block, and a single random value was drawn for each distinct block using seed 42. Blocks returning a value of 0.70 or higher formed the holdout set, and all samples within a block inherited the same assignment, thereby preventing adjacent observations sharing a block from being split across training and validation data. Because the procedure relies on a single block-level split, it is described as a spatial-block holdout rather than k-fold cross-validation.

2.7 Classifier comparison and production mapping

Three classifiers were benchmarked under the same spatial-block partition: Random Forest (RF), Classification and Regression Tree (CART), and Support Vector Machine (SVM). RF was configured with 500 trees and seed 42; CART used `maxNodes = 50` and `minLeafPopulation = 3`; and the SVM adopted a radial basis function kernel with `gamma = 0.05` and `cost = 10`. Because SVMs are sensitive to feature scale, SVM inputs were standardized using the mean and standard deviation computed from the training subset only, and the identical transform was then applied to the holdout subset. The model configurations are summarised in Table 2.

Table 2. *Classifier configurations evaluated in this study*

Classifier	Configuration
Random Forest	500 trees; random seed = 42
CART	<code>maxNodes = 50</code> ; <code>minLeafPopulation = 3</code>

Classifier	Configuration
SVM	RBF kernel; gamma = 0.05; cost = 10; training-only z-score standardization

Classifier performance was appraised through confusion matrices, overall accuracy and Cohen's kappa. RF was retained for the production workflow on the strength of its consistent use across multi-epoch mapping, forest-mask generation, sensitivity analysis and transition analysis. The single 2023-trained RF model was then applied to each of the six epoch composites, generating classified maps at 30 m resolution that were exported in UTM Zone 46N (EPSG:32646).

2.8 Independent reference data and accuracy assessment

A reference dataset of 200 locations was used for the 2023 accuracy assessment. Reference classes were assigned independently through visual interpretation of high-resolution imagery; the point locations, however, had been selected during an earlier classification iteration and were retained after the final model correction. Points were targeted at approximately 40 per class, and the realized per-class allocation was retained as sampled rather than forced to equality. The final RF prediction was then extracted at each site from the corrected 2023 classification and compared with the manual label.

Reference labels were assigned through visual interpretation of high-resolution imagery. Although the interpreter had access to the existing point file, the labels were based on image interpretation rather than automatically copied from the RF prediction; this procedure may nevertheless have introduced some awareness of the mapped class. The validation locations were also not re-drawn from the final map, and this sampling history is acknowledged as a limitation of the accuracy assessment.

2.9 Area-adjusted accuracy estimation

Class areas were computed in Google Earth Engine from geodesic pixel areas. The mapped-area proportions and the confusion matrix were passed to a Python implementation of the area-adjusted estimator of Olofsson et al. (2014), which assumes a stratified design in which the final-map classes serve as the strata. Because 27 of the 200 locations changed stratum after the final classifier correction (see below), this design assumption holds only approximately, and the resulting estimates are therefore reported as conditional. Each map class served as a stratum, and the weight of stratum i was taken as the mapped-area proportion of that class. For map stratum i and reference class j , the estimated area proportion takes the form:

$$\hat{p}_{ij} = \hat{W}_i \times (n_{ij} / n_{i\cdot}) \quad (6)$$

where $\hat{W}_i$ is the mapped-area proportion of stratum i , n_{ij} is the number of validation observations falling in cell (i,j) , and $n_{i\cdot}$ is the total number of observations in map stratum i . From this framework we derived area-adjusted overall, user's and producer's accuracies, standard errors, bias-corrected class-area estimates and 95% confidence intervals. The estimates were computed using final-map area weights applied to the retained reference sample. Because the reference locations were selected before the final classifier correction, and 27 of the 200 locations changed map-stratum assignment after that correction, these estimates and their confidence intervals should be interpreted as

conditional post hoc estimates rather than as strictly design-based inference from a newly drawn final-map stratified sample.

2.10 Forest-definition sensitivity analysis

To test how sensitive the reported forest extent is to the operational definition of Dense Forest, a sensitivity analysis was conducted for the 2023 classification only, repeating the classification under three NDVI thresholds: 0.45, 0.55 and 0.60. For each setting the class-label map was regenerated, the RF model was retrained under the identical sampling and spatial-block protocols, and the combined area of Dense Forest and Degraded Forest/Jhum was computed. The exercise therefore gauges the stability of mapped forest extent across plausible definitional choices; it does not amount to a separate accuracy assessment at each threshold.

2.11 Multi-temporal transition analysis

Transitions were examined between successive maps (1993–1998, 1998–2003, 2003–2008, 2008–2018 and 2018–2023), supplemented by a cumulative 1993–2023 matrix. Because the 2013 epoch was excluded (Section 2.2), the 2008–2018 interval spans ten years, so interval lengths are uneven (five, five, five, ten and five years). Annualised rates are therefore reported alongside the raw area changes rather than adjusting the interval structure. Each transition was encoded as:

$$T_{t,t+1} = C_t \times 10 + C_{t+1} \quad (7)$$

where C_t and C_{t+1} denote the initial and subsequent land-cover classes. Transition areas were obtained by multiplying the transition raster by `ee.Image.pixelArea()` and summing the product within the study boundary.

Gross forest loss was defined as a shift from Dense Forest or Degraded Forest/Jhum to Water, Agriculture/Settlement or Bare Land, and gross forest gain as a shift from any non-forest class to a forest class. A conversion from Dense Forest to Degraded Forest/Jhum was recorded as forest degradation rather than deforestation, whereas the reverse transition was treated as within-forest recovery or reclassification. Transitions involving Water were considered separately, since water-level fluctuations in the Kaptai Reservoir can produce changes unrelated to genuine land-cover conversion.

2.12 Data-quality control and reproducibility

For each observation year the workflow logged the number of available Landsat scenes, the mean scene-level cloud-cover metadata and the proportion of valid pixels surviving quality masking, allowing differences between epochs to be screened against image availability and post-mask coverage.

All image processing, classification, validation extraction, area computation and transition routines were implemented in the Google Earth Engine JavaScript API, with the area-adjusted accuracy calculations performed in Python. Dataset versions, sensor assignments, feature definitions and random seeds were recorded in the analysis scripts. The historical classifications

carry greater uncertainty than the 2023 assessment, since the independent reference observations pertain to the contemporary map and the validation locations were reused after the classifier correction.

2.13 Spectral separability diagnostic

To further probe the 2003–2008–2018 forest-composition swing (Section 3.4), a supplementary spectral-separability check was performed between the Dense Forest and Degraded Forest/Jhum classes at each epoch. For each epoch, NDVI values were sampled from pixels assigned to Dense Forest and Degraded Forest/Jhum in the classified map. A target of up to 500 randomly selected pixels per class was requested using seed = 42; the realized sample sizes varied according to the number of available pixels and are reported in Table A1. Class separability was quantified using the M-statistic, $M = |\mu_1 - \mu_2| / (\sigma_1 + \sigma_2)$, where $M > 1$ indicates stronger separation and $M < 1$ indicates greater spectral overlap. Results are reported in Section 4.4 and Table A1. Because this diagnostic used a supplementary composite-building routine that differs slightly from the production mapping workflow, its results are interpreted as exploratory evidence rather than as an independent accuracy assessment.

3. Results

This section reports the classifier selection, the independent accuracy assessment, and the resulting land-cover and change estimates. Throughout, a distinction is drawn between classifier performance, which describes how well the mapping algorithms reproduce the reference data, and ecological interpretation, which concerns what the mapped patterns may mean on the ground. The two are easily conflated in land-cover change work, and the accuracy figures in Sections 3.1 and 3.2 provide the context needed to read the transition results in Section 3.4 with appropriate caution.

3.1 Classifier comparison on the spatial-block holdout

Random Forest (RF), Classification and Regression Trees (CART), and Support Vector Machine (SVM) were each trained and evaluated on a common five-class candidate sample. The 750 candidate locations (150 per class) were partitioned by spatial block into 536 training observations and 214 spatially held-out observations, drawn from 68 distinct $0.1^\circ \times 0.1^\circ$ blocks of which 51 were assigned to training and 17 to holdout, following the protocols described in Sections 2.5–2.7. The spatially blocked design was used deliberately: when training and test pixels are drawn from overlapping or neighbouring locations, reported accuracy can be inflated by spatial autocorrelation rather than reflecting genuine generalization. Table 3 summarises the results.

Table 3. *Internal classifier comparison on the 214-sample spatial-block holdout*

Classifier	Overall accuracy	Kappa
SVM	78.04%	0.724
Random Forest	77.10%	0.712
CART	72.90%	0.659

SVM and RF performed nearly identically on the internal holdout, with SVM ahead by a slim 0.94 percentage points (78.04% versus 77.10%), and both clearly surpassed CART (72.90%). This small margin should not be over-read. Because no formal paired comparison or repeated spatial resampling was performed, the result is best interpreted as indicating comparable performance rather than a decisive advantage for SVM, particularly given the modest size of the 214-sample holdout. RF was nevertheless carried forward into production mapping, area estimation, the independent validation (Section 3.2), and the subsequent sensitivity and transition analyses (Sections 3.4–3.5). This choice reflects the documented competitiveness of RF with SVM in land-cover classification, its robustness to noisy reference labels, and its ready interpretability for a five-class problem. It is not a claim that RF is uniquely optimal; rather, it performed competitively with SVM and substantially better than CART, and its retention reflects practical consistency across the full analytical pipeline.

Note on a superseded internal figure. An earlier run under the same protocol produced an RF accuracy of 78.97% ($\kappa = 0.736$) under a coarser block-assignment scheme. The corrected scheme used here, which applies full $0.1^\circ \times 0.1^\circ$ spatial blocking with a single seeded per-block train/holdout draw (Section 2.6), yields the 77.10% ($\kappa = 0.712$) figure reported in Table 3. This is the figure carried into the manuscript; the earlier 78.97% figure is superseded and should not be cited elsewhere in the paper.

3.2 Independent accuracy assessment (area-adjusted, Olofsson et al., 2014)

Two accuracy figures arise from the 200-point reference set (Section 2.8), and both are reported for transparency. The raw (unweighted) accuracy, computed directly from the 200-point confusion matrix without correcting for the reference set's stratified sampling design, was **68.0% ($\kappa = 0.593$)**. The area-adjusted overall accuracy for the 2023 map, obtained with the estimator of Olofsson et al. (2014) by weighting each stratum by its mapped area rather than its sample count, was **73.17% (SE = 4.50 pp; 95% CI 64.34–82.00%)**.

These two values describe different quantities rather than competing estimates of a single one. The raw accuracy describes the observed reference sample, whereas the area-adjusted estimate accounts for the mapped-area proportions of the map strata and is therefore the primary estimate reported for the 2023 map (Section 2.8 explains the choice). The raw figure is presented alongside it for two reasons: it is the more conservative of the two and the one most often misquoted, and the roughly five-point difference between them, which a reader recomputing the confusion matrix by hand would notice, deserves an explicit explanation. Both figures fall below the 87% claimed, without independent validation, in the earlier thesis that this study supersedes.

The confusion matrix underlying both estimates is given in Table 4 (rows = map class, columns = reference class). Stratum sample sizes were Dense Forest 45, Degraded Forest/Jhum 53, Water 40, Agriculture/Settlement 38, and Bare Land 24, for a total of 200.

Table 4. *Reference-set confusion matrix (200 points)*

Map \ Reference	Dense Forest	Degraded Forest/Jhum	Water	Agriculture/Settlement	Bare Land
Dense Forest	35	8	0	2	0
Degraded Forest/Jhum	9	31	3	4	6
Water	0	0	40	0	0
Agriculture/Settlement	4	3	10	20	1
Bare Land	1	4	8	1	10

Per-class accuracies, computed as area-adjusted estimates with associated standard errors, are reported in Table 5.

Table 5. *Area-adjusted per-class user's and producer's accuracies*

Class	User's accuracy	UA SE	Producer's accuracy	PA SE
Dense Forest	77.78%	6.27	93.56%	1.36
Degraded Forest/Jhum	58.49%	6.83	43.50%	10.73
Water	100.00%	0.00	60.72%	8.59
Agriculture/Settlement	52.63%	8.21	41.76%	16.03
Bare Land	41.67%	10.28	33.60%	12.23

Per-class accuracy reveals a clear asymmetry that matters for how the change results should be read. Dense Forest and Water are mapped reliably, although in different senses. Water attains a perfect 100% user's accuracy but a producer's accuracy of only 60.7%: every pixel the map labels as water genuinely is water, yet roughly 40% of true water pixels are assigned to another class. This pattern may partly reflect seasonal or shoreline-related variation near the reservoir (cf. Karmakar et al., 2019), although that explanation was not directly tested and is considered further in the Discussion. The three classes that carry the interpretive weight in any forest-degradation narrative, Degraded Forest/Jhum, Agriculture/Settlement, and Bare Land, are markedly weaker, with user's accuracies of 42–58% and producer's accuracies of 34–44%. Because these per-class figures are area-adjusted estimates rather than simple ratios of the raw matrix, they should not be expected to reconcile exactly with Table 5. The implication for Section 3.4 is direct: changes involving the weaker classes carry substantially more error than those involving the well-mapped classes, and transition magnitudes among them should be interpreted accordingly.

Area-adjusted class areas are given in Table 6. The total study area is 576,534.38 ha.

Table 6. *Area-adjusted class areas with 95% confidence intervals*

Class	Mapped (ha)	Adjusted (ha)	± 95% CI (ha)
Dense Forest	396,367	329,516	49,877
Degraded Forest/Jhum	99,640	133,985	46,877
Water	30,624	50,436	8,491
Agriculture/Settlement	35,084	44,219	25,827
Bare Land	14,820	18,378	9,264

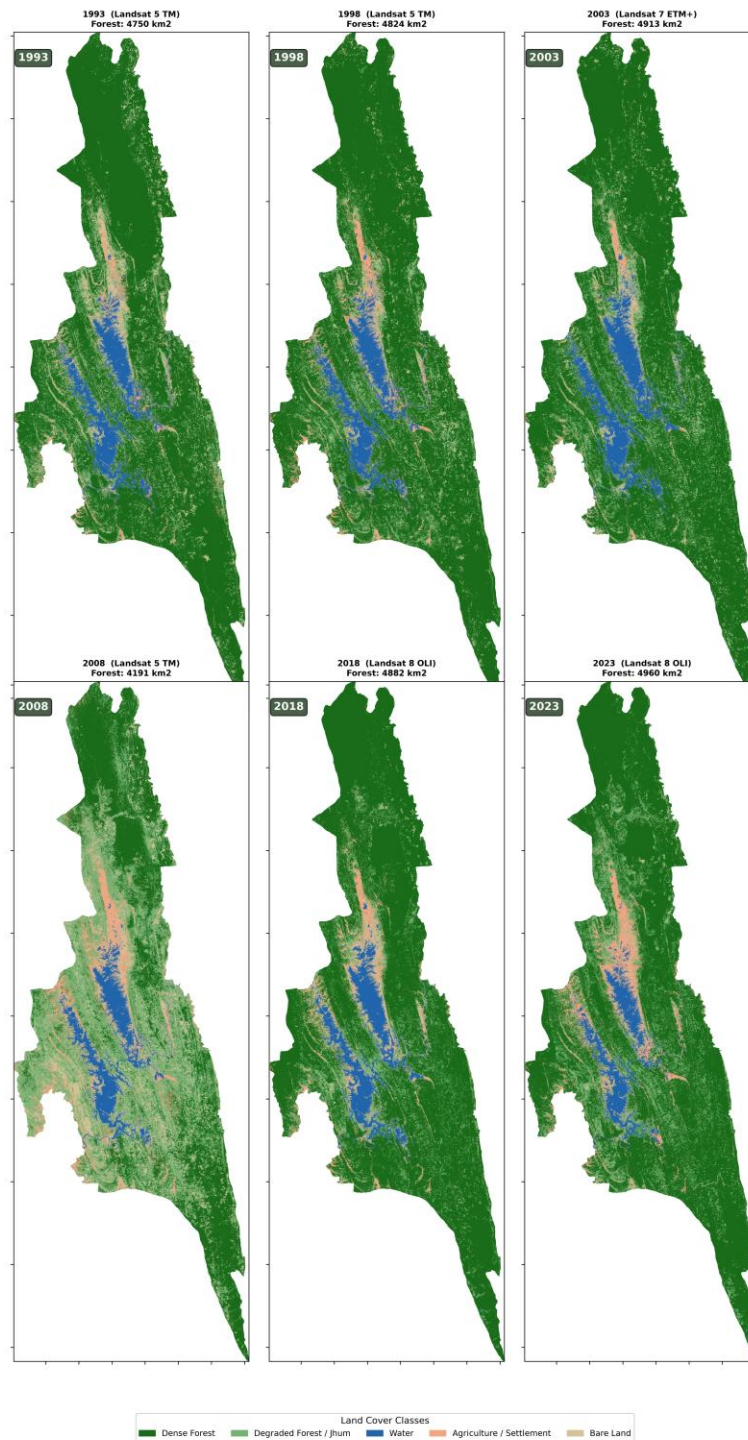
The area-adjusted confidence intervals are wide relative to their point estimates; for Degraded Forest/Jhum, for example, the interval ($\pm 46,877$ ha) is roughly 35% of its own estimate. Any downstream area or change figure in this paper should therefore be read with this uncertainty attached rather than treated as an exact quantity.

3.3 Per-epoch land-cover composition

Table 7 reports the classified extent of each land-cover class in every epoch. All areas are in hectares, and the per-epoch totals close exactly to the study-area extent, serving as an internal check on the area-computation pipeline. Two features stand out. First, the balance between Dense Forest and Degraded Forest/Jhum shifts markedly across the record, with a large transfer of area between the two around 2008 that is examined in Section 3.4. Second, the combined forest extent (Dense Forest plus Degraded Forest/Jhum) remains comparatively stable. Together with the sensitivity analysis in Section 3.5, this suggests that much of the apparent volatility reflects reclassification within the forest category rather than movement across the forest–non-forest boundary. It bears repeating that these extents are map-based and therefore inherit the classification error discussed in Section 3.2; they are not independent of it. They are also mapped (unadjusted) areas, because the independent reference assessment was performed only for the 2023 map; no area-adjusted correction is available for the historical epochs.

The six-epoch sequence is shown in Figure 2.

Temporal Land Cover Change - Rangamati District (1993-2023)
Random Forest Classification | 2013 Excluded (Insufficient L8 Data)



All maps: RF (500 trees) trained on 2023 features applied to all epochs (temporal transfer). Projection: UTM Zone 48N.

Figure 2. Six-epoch land-cover classification of Rangamati District, 1993–2023, with mapped forest area annotated per panel. All forest areas are mapped (unadjusted) estimates from the single 2023-trained Random Forest applied to each epoch; they are not area-adjusted and are not independently validated for historical epochs.

Table 7. Per-epoch land-cover composition (ha)

Class	1993	1998	2003	2008	2018	2023
Dense Forest	404,602	414,304	416,402	202,530	414,183	396,367
Degraded Forest/Jhum	70,711	68,116	75,242	216,613	74,277	99,640
Water	41,609	40,750	47,169	38,485	41,346	30,624
Agriculture/Settlement	18,081	25,425	10,755	42,588	23,220	35,084
Bare Land	41,531	27,938	26,967	76,318	23,508	14,820
Total	576,534	576,534	576,534	576,534	576,534	576,534

The 2023 production map is shown in Figure 3.

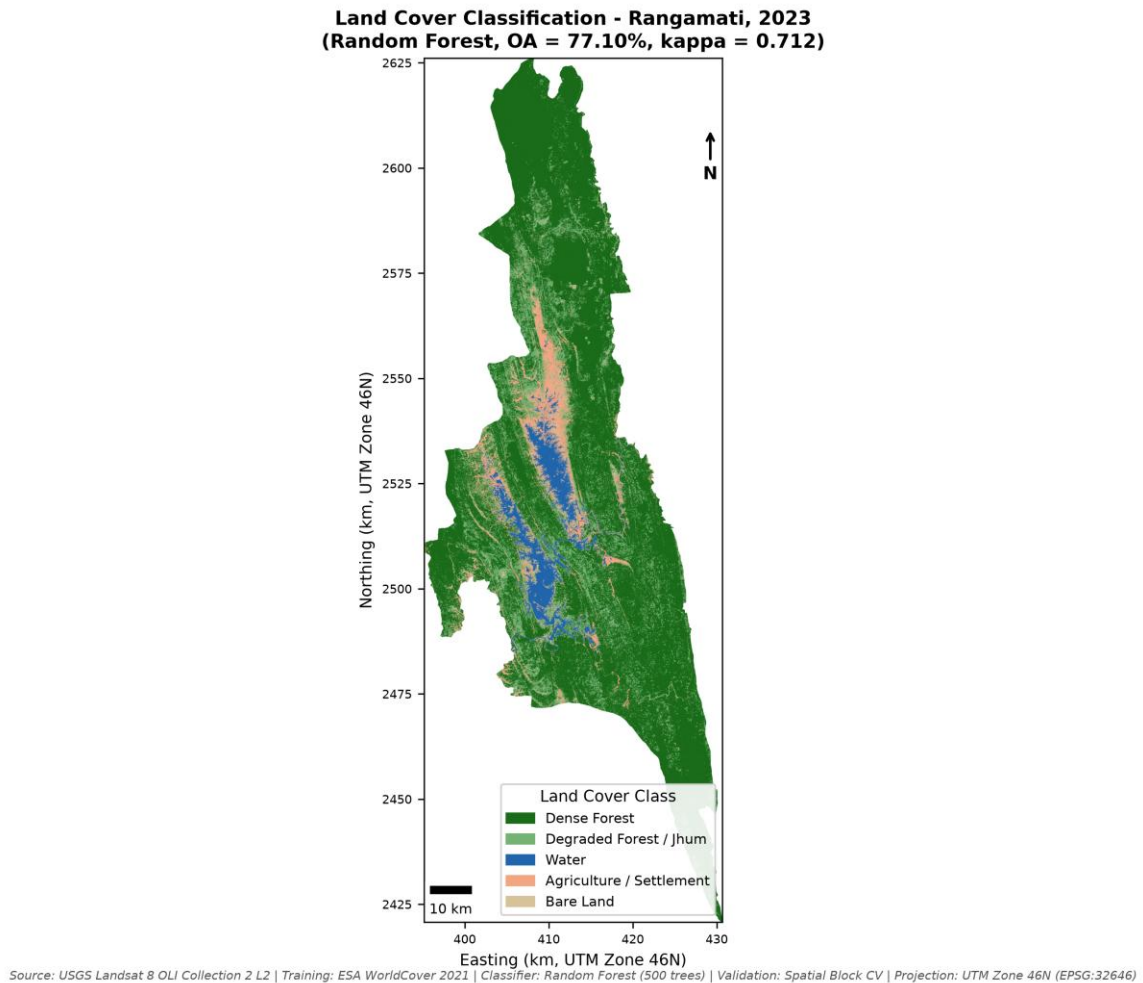


Figure 3. *Land-cover classification of Rangamati District, 2023, produced with the 2023-trained Random Forest. The figure title reports the internal spatial-block holdout accuracy ($OA = 77.10\%$, $\kappa = 0.712$; Table 3); the independent area-adjusted overall accuracy for this map is reported in Section 3.2.*

3.4 Multi-decadal transition analysis

Transition areas were derived from classified-map comparisons and were not corrected using an error-adjusted transition estimator. They are therefore mapped estimates whose uncertainty reflects the per-class errors reported in Section 3.2. Net forest change by interval, expressed against the combined Dense Forest plus Degraded Forest/Jhum baseline, is summarised in Table 8.

The 30-year comparison is shown in Figure 4.

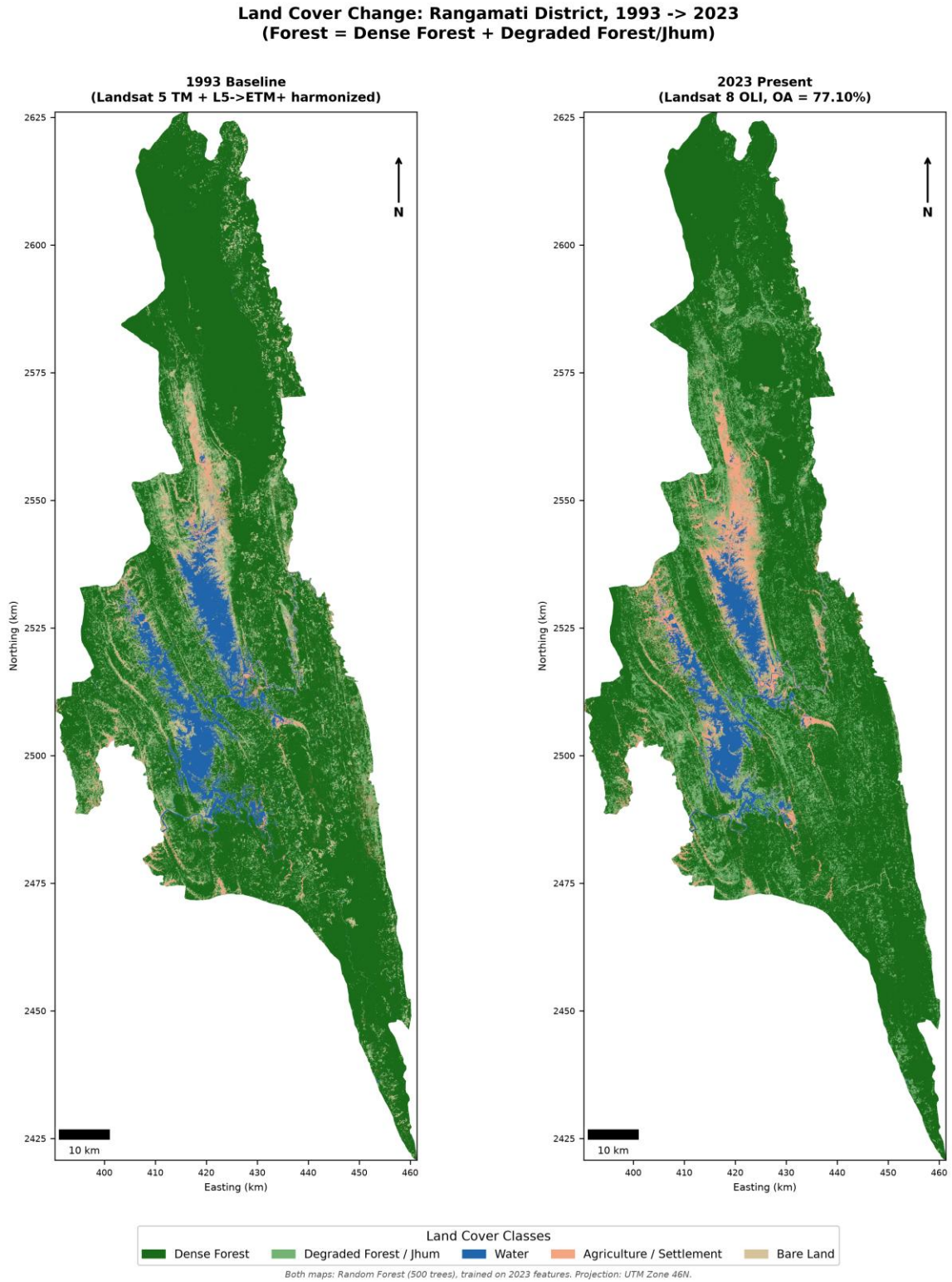


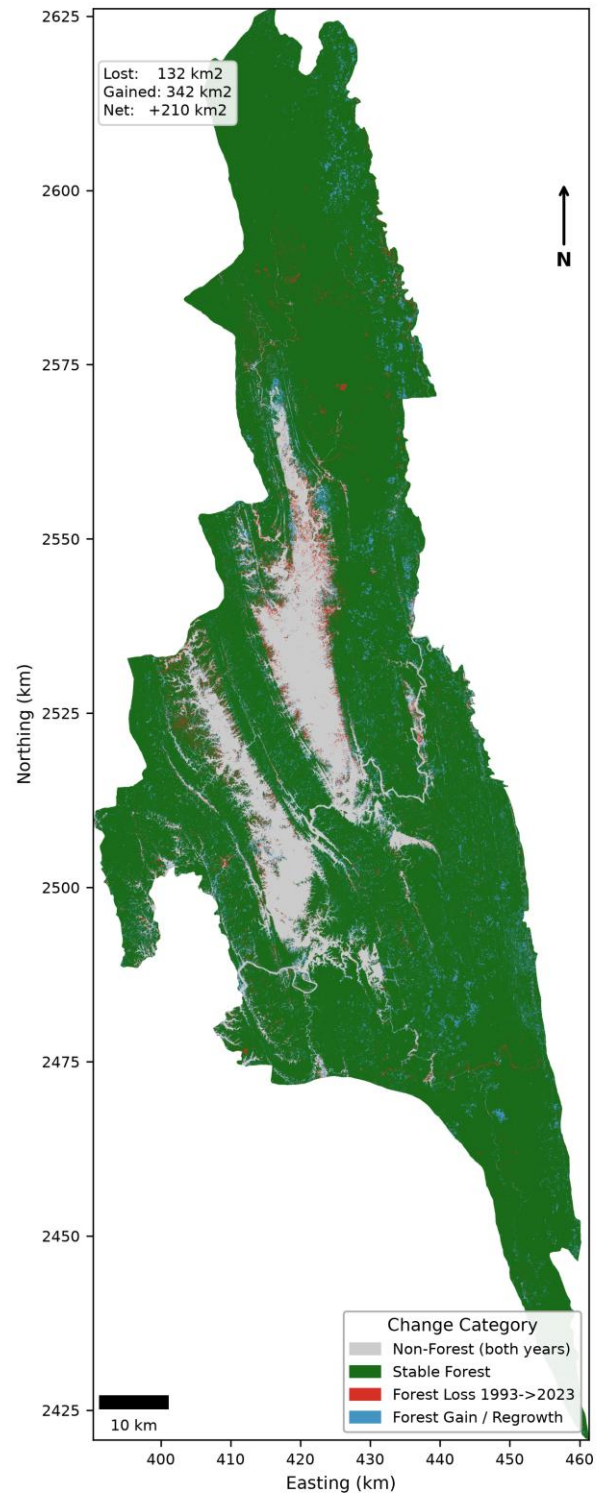
Figure 4. Comparison of mapped land-cover classification for 1993 and 2023. Areas are mapped (unadjusted) estimates from the 2023-trained Random Forest applied to each epoch; only the 2023 map has an independent area-adjusted accuracy assessment (Section 3.2).

Table 8. *Multi-decadal forest transition by interval*

Interval	Net forest change	Annualized rate
1993–1998	+7,107 ha	+1,421.5 ha/yr
1998–2003	+9,223 ha	+1,844.6 ha/yr
2003–2008	–72,501 ha	–14,500.2 ha/yr
2008–2018 (10 yr)	+69,318 ha	+6,931.8 ha/yr
2018–2023	+7,547 ha	+1,509.3 ha/yr
Cumulative 1993–2023	+20,693 ha	+689.8 ha/yr

The spatial pattern of the cumulative 1993–2023 forest change is shown in Figure 5.

Forest Cover Change - Rangamati District, 1993-2023
(Forest = Dense Forest + Degraded Forest/Jhum)



Pixel-level change detection at 30m resolution. Forest = Dense Forest (NDVI >= 0.55) + Degraded Forest/Jhum (NDVI 0.25-0.55). Projection: UTM Zone 46N.

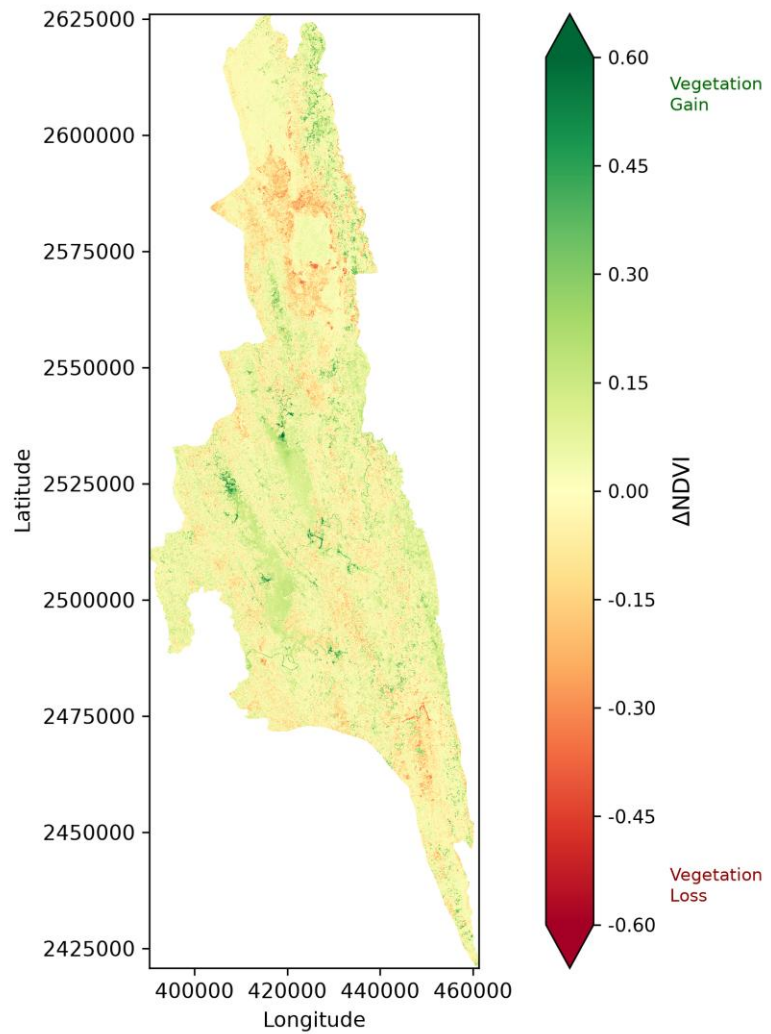
Figure 5. *Forest cover change in Rangamati District, 1993–2023, showing stable forest, forest loss, and forest gain with summary statistics. Change areas are mapped (unadjusted) estimates derived from the classified maps and are not error-adjusted; the 2003–2008–2018 swing is an open finding (Sections 3.4 and 4.4), not a confirmed forest-loss or recovery event.*

Over the full 30-year record the study area shows a net forest gain of 20,693 ha, equivalent to +4.4% relative to the combined 1993 Dense Forest + Degraded Forest/Jhum baseline of 475,313 ha, rather than the loss reported by the original, uncorrected thesis. That headline figure must be read with one overriding caveat: it is driven almost entirely by a single large swing centred on 2003–2018, in which Dense Forest falls from 416,402 ha (2003) to 202,530 ha (2008) and then recovers to 414,183 ha by 2018. Considered on its own, Dense Forest shows a small net decline over the full 30 years (404,602 ha in 1993 to 396,367 ha in 2023, –8,235 ha, or –274 ha/yr), alongside substantial two-way churn against the Degraded Forest/Jhum class (Dense→Degraded 60,540 ha cumulative; Degraded→Dense 38,706 ha cumulative). Taken together, the record reads more as forest-quality churn than as outright forest loss, though this interpretation is conditional on the accuracy caveats in Section 3.2.

A dedicated diagnostic (scene count, mean scene-level cloud-cover metadata, and post-mask valid-pixel fraction for all six epochs) was run to test whether input-image quality could explain the 2003–2008–2018 swing. The 2008 epoch (23 scenes, 17.9% mean cloud cover, 100% valid-pixel coverage) was not an outlier against any of the other five epochs (23–37 scenes, 12.3–34.6% cloud cover, and 100% valid-pixel coverage in every epoch, including 1998 at 34.6% cloud cover with no comparable swing). The diagnostic therefore did not indicate a gross missing-data or coverage problem, but it cannot exclude residual sensor, radiometric, compositing, or class-boundary effects as contributors to the observed swing. The swing is thus reported as an open finding rather than a data artefact or a confirmed ecological signal; possible contributors include residual cross-sensor differences (Landsat 7-based 2003, Landsat 5-based 2008, and Landsat 8-based 2018 composites), compositing effects, and instability in the Dense Forest / Degraded Forest/Jhum class boundary, which the present analysis does not permit to be separated. This is framed in the Discussion as a direction for further investigation rather than asserted as a confirmed collapse-and-recovery cycle.

The NDVI change between 2023 and 1993 (Figure 6) and the forest-cover trend (Figure 7) summarise the pattern.

NDVI Change (2023 – 1993), Rangamati District
After Roy et al. (2016) cross-sensor harmonization



Positive values indicate greening; this map is descriptive and does not represent a validated change-detection product.

Figure 6. Change in the Normalized Difference Vegetation Index (NDVI) between 2023 and 1993 (Δ NDVI), shown with a diverging colour scale. The map supports interpretation of Dense↔Degraded reclassification and is not a standalone classification or accuracy figure.

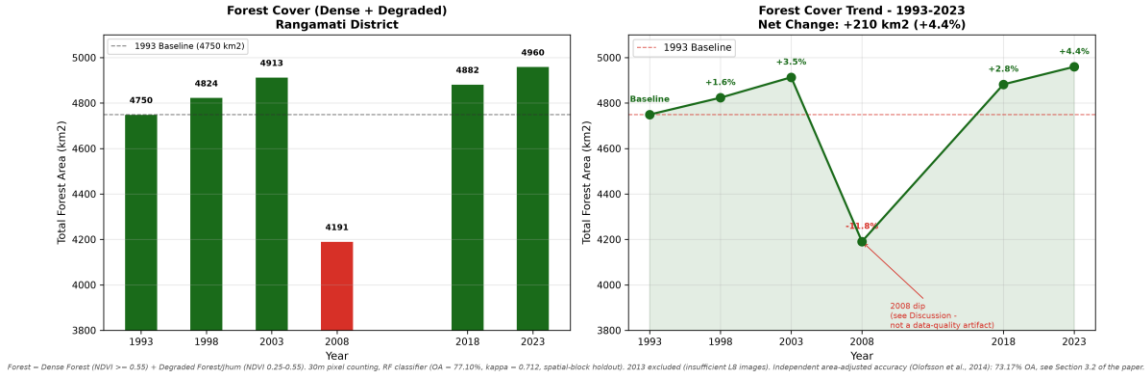


Figure 7. Forest cover trend for Rangamati District, 1993–2023, as a bar and line chart, with the 2008 reduction annotated. Forest areas are mapped (unadjusted) estimates from the classified maps; the 2008 reduction and the surrounding swing are treated as an open finding (Sections 3.4 and 4.4), not a confirmed ecological event.

3.5 Forest-definition sensitivity analysis

The Dense Forest / Degraded Forest/Jhum boundary defined in Section 2.4 and evaluated in Section 2.10 was tested at three NDVI thresholds, 0.45, 0.55 (operational), and 0.60, to quantify how much the 2023 area figures depend on this single parameter choice (Table 9).

Table 9. Sensitivity of forest area to the NDVI forest-definition threshold

NDVI threshold	Dense Forest area (ha)	Total Forest area (ha) ¹
0.45 (loose)	426,497	494,515
0.55 (operational)	396,367	496,007
0.60 (strict)	318,658	492,311

¹ Total Forest = Dense Forest + Degraded Forest/Jhum combined.

The Dense Forest / Degraded Forest/Jhum split within total forest is highly sensitive to this threshold: Dense Forest area ranges from 318,658 to 426,497 ha across the tested range, a spread of 107,839 ha, or roughly 27% of the operational estimate. Total Forest area, by contrast, is essentially stable, varying only from 492,311 to 496,007 ha, a spread of 3,696 ha, or about 0.7%, as the threshold moves. This pattern confirms that the threshold mainly reclassifies pixels *within* the forest category, between "dense" and "degraded/Jhum", rather than moving pixels across the forest/non-forest boundary. It indicates that the estimated 2023 total forest extent is comparatively insensitive to the tested threshold. It does not establish robustness of historical area trajectories or interval-specific transition estimates, which were not recalculated under each threshold; any Dense-vs-Degraded split, including the 2003–2008–2018 swing in Section 3.4, should accordingly be read with this caveat attached. This analysis addresses only the NDVI threshold; it does not test sensitivity to other components of the pipeline, such as the classification model or compositing scheme, so its reassuring conclusion should not be generalised beyond this parameter.

4. Discussion

4.1 Principal findings

Reconstructing land cover across six Landsat epochs produced a district-scale result that departs from the conventional deforestation narrative for the Chittagong Hill Tracts: combined forest extent (Dense Forest plus Degraded Forest/Jhum) remained broadly stable over 1993–2023, with a net increase of 20,693 ha (+4.4% of the 1993 baseline). This should not be read as large-scale forest expansion. The more defensible reading is that total forested area neither collapsed nor recovered dramatically, while its composition shifted substantially. Dense Forest declined modestly over the record (−8,235 ha; −274 ha yr^{−1}), and there was pronounced two-way exchange with the Degraded Forest/Jhum class. The pattern is therefore more consistent with movement of area between the two forest-related classes than with a monotonic loss trajectory.

That reading is reinforced by the forest-definition sensitivity test (§3.5): varying the Dense Forest NDVI threshold from 0.45 to 0.60 changed Dense Forest area by roughly 27% but total forest extent by only about 0.7%. Much of the apparent volatility therefore reflects reclassification within the forest category rather than conversion across the forest–non-forest boundary.

4.2 Comparison with previous research

Direct comparison with earlier regional work is limited by differences in class schemes, sensors, periods, and validation design; the comparisons below are therefore cautious and not treated as contradictions.

The closest analogue is Chakma et al. (2023), who mapped land cover across the Hill Tracts for 1998, 2008, and 2018 with object-based classification and reported a forest-cover increase of about 58% over 1998–2018. Both studies find net forest gain rather than the classic loss narrative, but the magnitudes differ greatly. Their four-class scheme aggregates into a single "forest" category the cover types we separate into Dense Forest and Degraded Forest/Jhum, so much of what registers as "forest gain" under a lumped scheme appears here as internal Dense↔Degraded exchange. Their study also covers the entire Hill Tracts over a shorter period using a different method. These are related but distinct measurements, not conflicting findings.

At protected-area scale, Hossain et al. (2023) reported roughly 16% forest loss in the Fashiakhali Wildlife Sanctuary impact area between 1994 and 2021. The contrast with both our district-scale stability and Chakma et al.'s (2023) gain shows that forest trajectories within the Hill Tracts are strongly sub-region-specific: a district aggregate can remain stable while individual reserves lose forest and other tracts regrow. This is consistent with shifting-cultivation dynamics in which fallow regrowth and re-clearing coexist, and it cautions against reading any single district-level net figure as a uniform on-the-ground trend.

4.3 Methodological interpretation

The classifier comparison (§3.1) supports a bounded conclusion. On the spatially blocked holdout, SVM and Random Forest were effectively equivalent (78.04% versus 77.10%), with CART clearly

behind (72.90%); the 0.94-point SVM–RF gap, measured on a single 214-sample holdout without a paired test, does not establish a meaningful advantage. This near-equivalence is consistent with broader comparative evidence, and carrying RF forward rested on pipeline consistency and interpretability rather than statistical superiority.

The spatial-block holdout is a deliberately conservative choice. Random splits of spatially autocorrelated data tend to inflate reported accuracy relative to a spatially independent split. The block design used here therefore yields a more conservative estimate under these conditions and is not directly comparable with results from random or resubstitution-based validation.

The area-adjusted assessment is the most consequential element for reading the numbers, because it separates four non-interchangeable quantities: map accuracy, mapped area, area-adjusted area, and ecological meaning. The gap between the raw (68.0%) and area-adjusted (73.17%) overall accuracies illustrates the point. Two limitations qualify these figures. First, the area-adjusted assessment applies only to the 2023 map; historical areas (§3.3) and transitions (§3.4) are mapped, unadjusted quantities, and the 73.17% figure should not be read backward as a measure of historical-map reliability. Second, the estimates are conditional rather than strictly design-based: the 200 reference locations were selected before the final classifier correction, 27 changed map-stratum assignment after it, and the interpreter had access to the existing point file, so the labels, though image-based, were not guaranteed blind to the classifier output.

Cross-sensor harmonization with the coefficients of Roy et al. (2016) reduces systematic offsets among Landsat 5, 7, and 8 but does not remove all inter-sensor differences. The WorldCover-derived training labels add a further, well-characterized limitation: WorldCover has roughly 75% overall accuracy with meaningful local disagreement and depicts approximately the 2021 condition. Combined with the fixed 2023 NDVI rule, a location whose true class differed from its 2021/2023 state contributes a label that may not match its historical condition, a mismatch most consequential for the earliest epochs and the central reason the historical maps are framed as model-based reconstructions under a common labelling framework rather than as independently validated maps.

4.4 Interpretation of forest-change patterns

The transition results (§3.4) must be read through the per-class accuracies (§3.2). The classes carrying the most interpretive weight, Degraded Forest/Jhum, Agriculture/Settlement, and Bare Land, are precisely those the map estimates least reliably (user's accuracies 42–58%; producer's accuracies 34–44%), and the area-adjusted confidence intervals are wide (for Degraded Forest/Jhum, roughly 35% of the point estimate). Degraded and transitional forest cover, including shifting-cultivation mosaics, is intrinsically hard to separate from spectral neighbours, and the Degraded Forest/Jhum class deliberately spans a broad, heterogeneous range. The large Dense↔Degraded exchange is therefore best read as a combination of genuine landscape change and classification instability at the class boundary; degradation and within-forest recovery are recorded as defined but not ground-verified at the reported magnitude.

The 2003–2008–2018 swing is reported as unresolved. A targeted diagnostic showed the 2008 epoch was not an outlier in scene count, mean scene-level cloud-cover, or post-mask valid-pixel

fraction, ruling out a gross missing-data or coverage failure. It does not exclude residual cross-sensor differences among the Landsat 7-based 2003, Landsat 5-based 2008, and Landsat 8-based 2018 composites, compositing effects, or instability at the Dense Forest / Degraded Forest/Jhum boundary. The analysis cannot separate these contributors, and none is demonstrated, so the swing is treated as an open signal rather than a confirmed collapse-and-recovery cycle.

A spectral-separability diagnostic, based on the M-statistic, was run across all six epochs to test whether the swing coincides with weaker separability between Dense Forest and Degraded/Jhum in the classified NDVI feature space. The result does not identify 2008 as a unique outlier: 1998 ($M = 0.757$) and 2008 ($M = 0.921$) both show comparatively weak separation, while 1993, 2018, and 2023 are clearly well separated ($M > 1.14$; Table A1). Both 1998 and 2008 are Landsat-5-based composites, which is at least suggestive of a sensor-specific effect, although 1993, also Landsat-5-based, is well separated ($M = 1.173$), so this explanation remains only partially supported. The diagnostic therefore neither confirms nor rules out class-boundary instability as a contributor to the swing; it is reported as inconclusive rather than resolved. Two further limitations bound the transition results: they are mapped estimates, not corrected with an error-adjusted transition estimator, so interval-specific magnitudes are approximate and class-level uncertainties propagate directly into them.

4.5 Implications and contribution

The realistic contribution is methodological and regional rather than a definitive statement about Rangamati's forest trajectory. The study demonstrates that a fully free and reproducible pipeline, combining Landsat Collection 2 surface reflectance, Earth Engine processing, WorldCover-derived labels, and a Python implementation of the Olofsson estimator, can produce a multi-decadal, multi-class, independently assessed land-cover record for a data-scarce hill district, with uncertainty reported transparently. Individually, these are established tools rather than new methods; what this study adds is combining them into one transparent, end-to-end workflow for a hill district that otherwise lacks a validated multi-decadal record, and presenting the inconvenient findings (Section 3.4) with the same weight as the favourable ones.

Management implications should stay proportional to this evidence. The results are consistent with district-scale forest persistence accompanied by substantial internal turnover, which aligns qualitatively with reports that community forestry has supported regrowth in parts of the Hill Tracts even as specific reserves lose forest. The analysis does not identify causes, quantify policy effectiveness, or confirm ecological recovery, and no such claim is made. The most defensible practical message is that monitoring in this landscape should track the dense-versus-degraded distinction and the forest-quality churn it captures, not only net forest area, since a stable aggregate can conceal considerable compositional change.

5. Conclusion

This study reconstructed land cover in Rangamati District across six Landsat epochs from 1993 to 2023 using five operational remote-sensing classes and a common multi-sensor processing framework. The classifier comparison showed similar spatial-block performance for Random Forest and support vector machine, and Random Forest was retained for production mapping. The

manually interpreted 2023 reference assessment yielded an area-adjusted overall accuracy of 73.17% (95% confidence interval: 64.34–82.00%). The mapped record indicates that combined forest extent was broadly stable, increasing by 20,693 ha, or 4.4% relative to the 1993 baseline, while mapped Dense Forest declined by 8,235 ha. These findings distinguish aggregate forest extent from the distribution of the operational Dense Forest and Degraded Forest/Jhum classes. These classes are operational remote-sensing proxies and should not be interpreted as direct field measurements of forest condition.

The sensitivity analysis reinforces this interpretation. Changing the NDVI threshold altered total mapped forest area by 0.7% but changed the Dense Forest estimate by 27%, indicating that the threshold chiefly redistributes area between the two forest-related classes. The pronounced 2003–2008–2018 redistribution therefore remains unresolved and cannot be interpreted as confirmed forest loss, recovery, or degradation at the reported magnitude. Area-adjusted validation applies only to the 2023 map, whereas historical classifications and transition estimates remain unadjusted mapped estimates without equivalent reference-point validation. For tropical hill landscapes with shifting cultivation and heterogeneous vegetation, monitoring should consequently report total forest extent alongside the uncertainty and redistribution associated with operational forest classes.

Appendix A

Table A1. *Spectral separability (M-statistic) between Dense Forest and Degraded/Jhum, by epoch.*

Epoch	M-statistic	Mean NDVI (Dense)	SD (Dense)	Mean NDVI (Degraded)	SD (Degraded)	n (Dense)	n (Degraded)
1993	1.173	0.660	0.062	0.514	0.062	334	71
1998	0.757	0.620	0.097	0.485	0.081	346	66
2003	0.999	0.640	0.082	0.478	0.079	347	71
2008	0.921	0.591	0.060	0.480	0.061	171	195
2018	1.270	0.698	0.059	0.564	0.046	344	71
2023	1.148	0.688	0.063	0.558	0.050	334	90

M-statistic = $|\text{mean1} - \text{mean2}| / (\text{SD1} + \text{SD2})$. $M > 1$ indicates well-separated classes; $M < 1$ indicates substantial spectral overlap. NDVI sampled from n randomly selected pixels per class per epoch, stratified by the epoch's classified map.

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Appendix B

Reproducibility and Repository

The complete Google Earth Engine scripts, Python accuracy workflow, validation data, reproducibility guide, and publication notebook are available at:

<https://github.com/ShohanRony/rangamati-landcover-change-1993-2023>

The repository documents the required Earth Engine assets, execution order, classifier settings, area-adjusted accuracy workflow, known limitations, and the distinction between validated 2023 estimates and unadjusted historical reconstructions.